

Fusion Training for Mathematical Generalization in Large Language Models

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Background

LLMs need both long-form reasoning for hard problems and concise answers for simple queries.

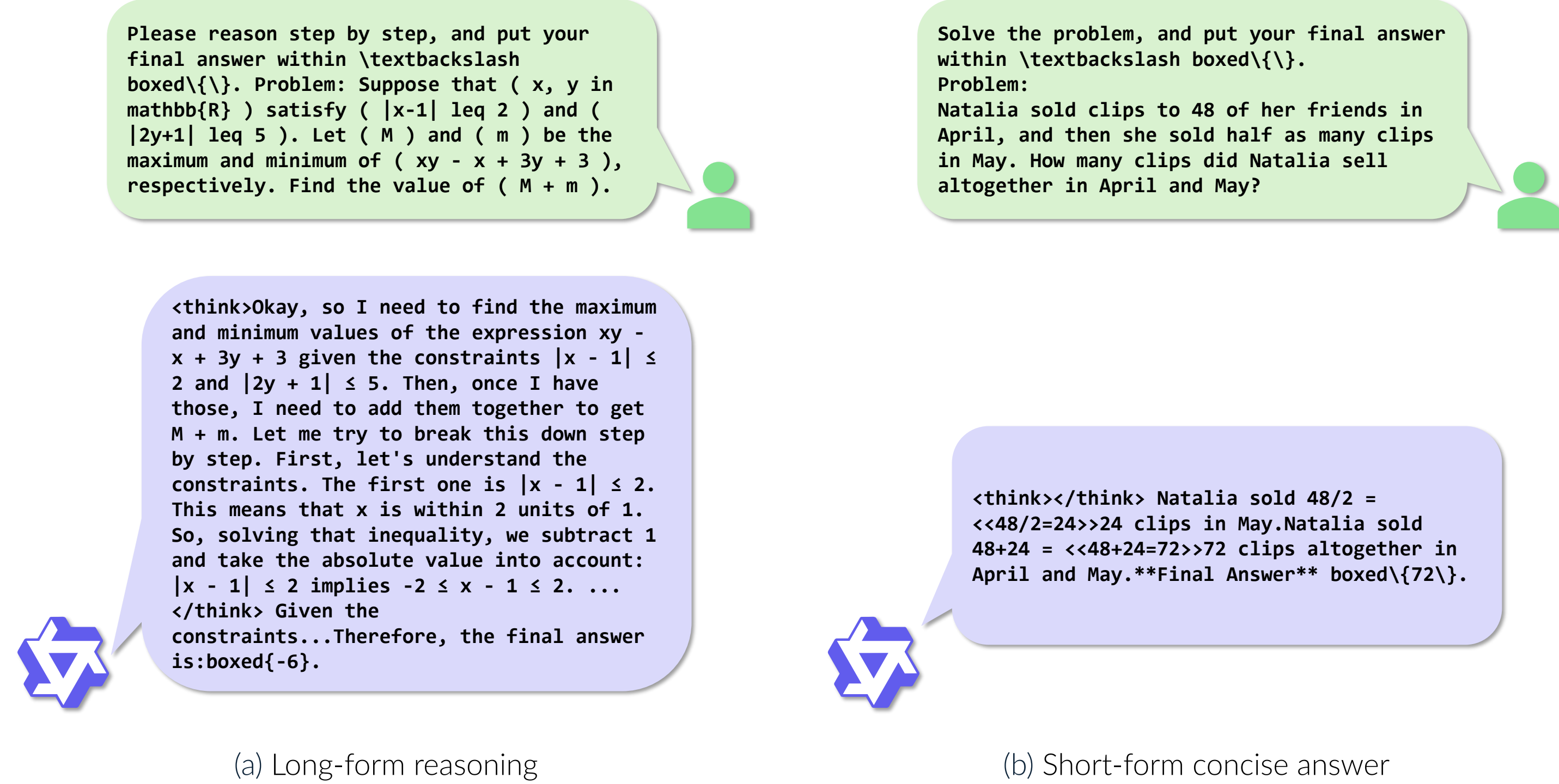


Figure. Examples of long-form reasoning and short-form concise answers.

Motivation

Thinking Mode Fusion (TMF) puts both modes in one model, but its training dynamics remain under-explored.

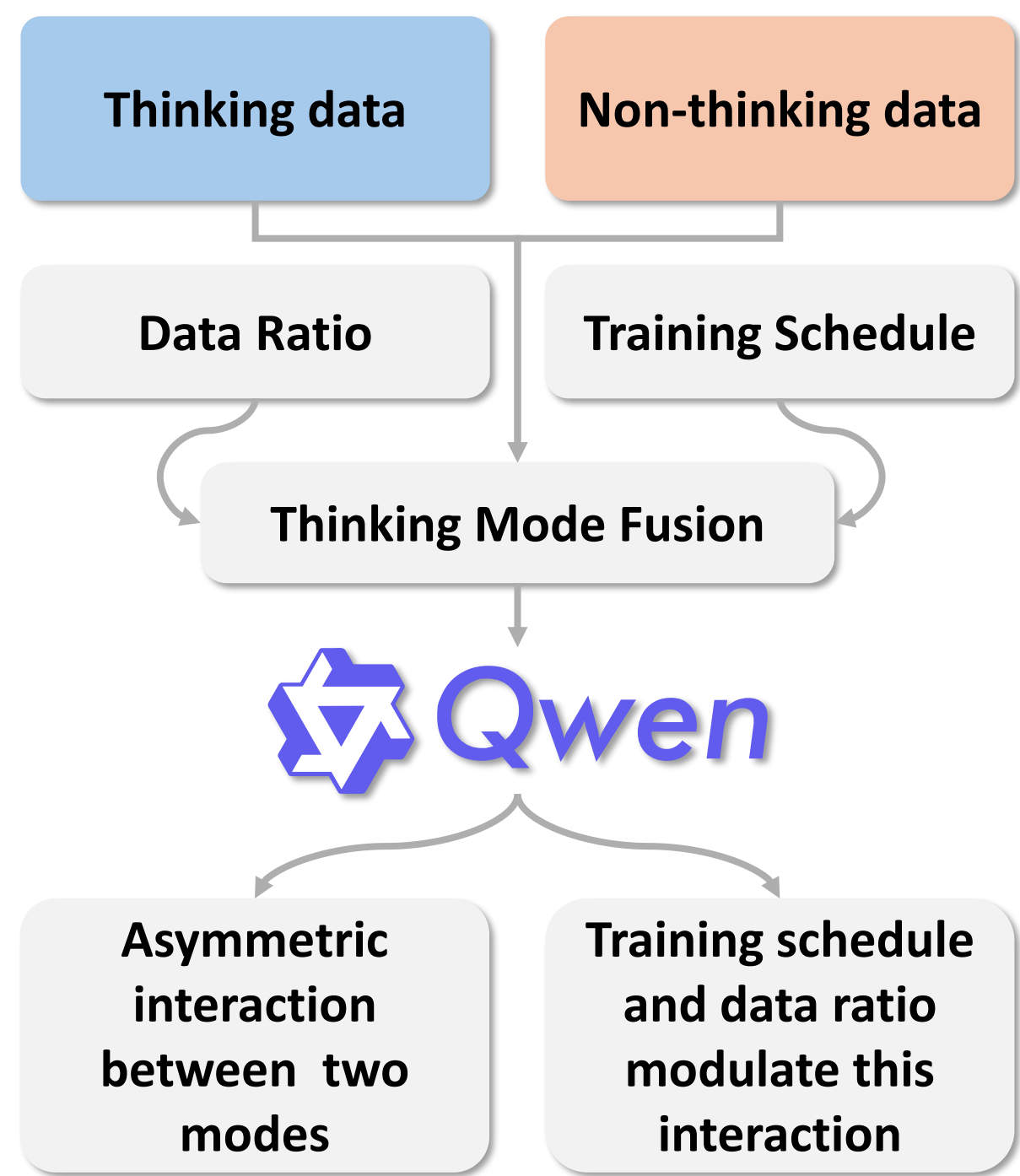


Figure. An overview of the experimental design for analyzing TMF under different training orders and data ratios.

Key Questions

- ★ RQ1: Do the non-thinking and thinking modes affect each other's performance?
- ★ RQ2: How does the different training schedule affect Thinking Mode Fusion?
- ★ RQ3: How does the different data ratio affect Thinking Mode Fusion?
- ★ RQ4: What is the quantitative relationship between the non-thinking and thinking modes?

Overview of Thinking Mode Fusion Training

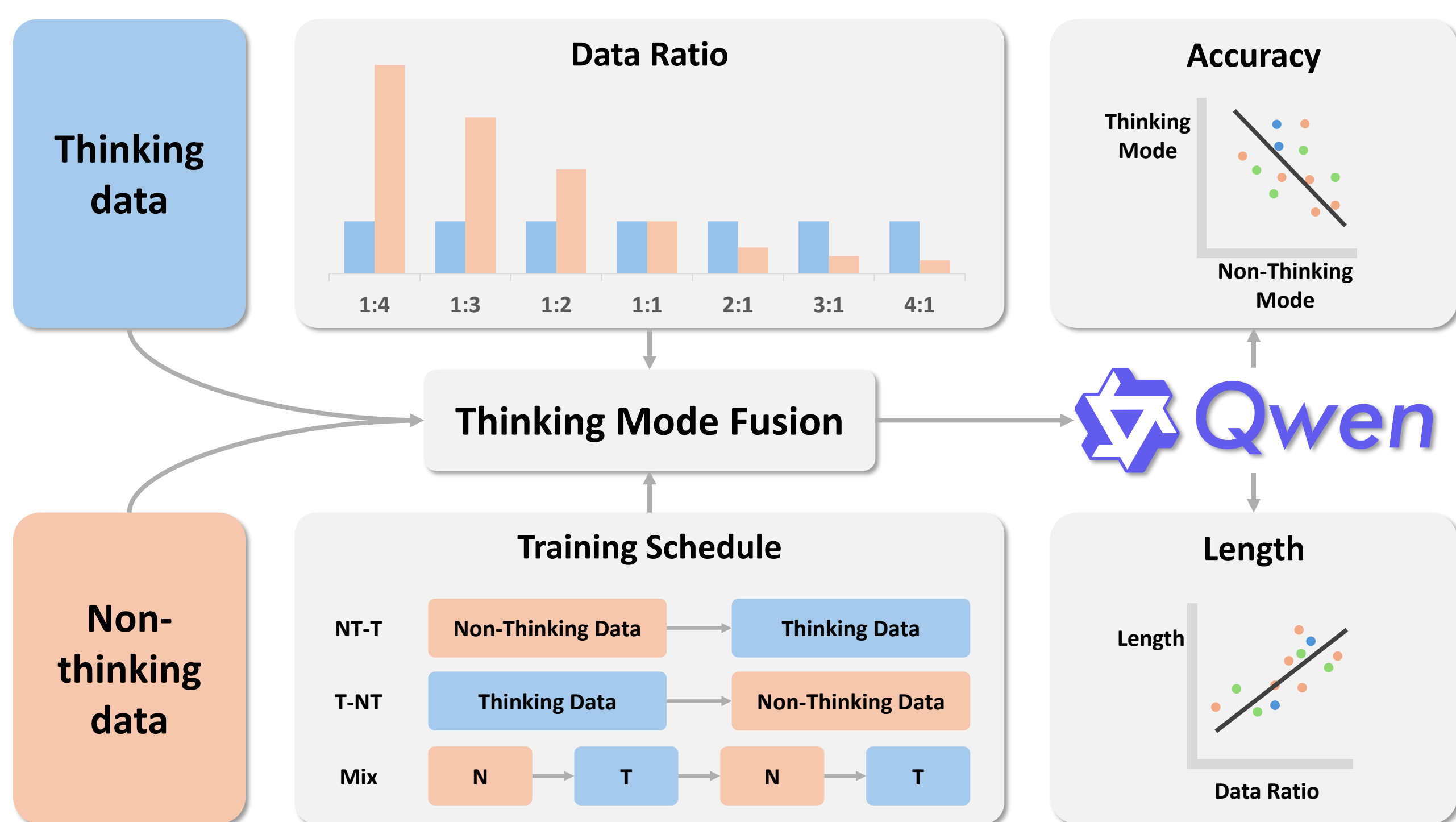


Figure. We construct two datasets: thinking mode and non-thinking mode. We vary the data ratio (T:NT from 1:4 to 4:1) and schedule the resulting SFT stream using three strategies: T-NT (first thinking then non-thinking), NT-T (non-thinking then thinking), and Mix (interleaved to match the target ratio). Each configuration fine-tunes Qwen3 model and is evaluated on both thinking and non-thinking benchmarks to quantify interference and trade-offs.

Fusion Bench

Mode	Ratio (T : NT)	Count	Mean	Min	Max
Non-thinking	4 : 1	375	671	330	1,540
	3 : 1	500	672	330	1,540
	2 : 1	750	681	330	1,540
	1 : 1	1,500	692	298	1,817
	1 : 2	3,000	692	298	1,817
	1 : 3	4,500	695	298	1,863
	1 : 4	6,000	694	298	1,863
	-	1,500	11,877	1,903	54,786
Thinking	-	-	-	-	-

Fusion Bench. **T** denotes thinking mode data and **NT** denotes non-thinking mode data. We fix the thinking mode subset size ($N_T = 1500$) and vary the non-thinking subset size (N_{NT}) accordingly. Mean denotes the mean length of the reasoning traces.

RQ1: Interaction between Non-Thinking and Thinking Mode

Table. Performance comparison of accuracy and reasoning length under different data ratios and training schedule settings. We report Len (average response length) and Acc (accuracy) for three runs. The rightmost Mean column averages results across all data ratios, while the Mean row in each mode block averages over training schedules. The Mix training schedule achieves the highest overall accuracy, while increasing the thinking data ratio consistently improves thinking-mode performance and leads to longer responses.

		Data Ratio (T:NT, T=1500)																	
Mode	Setting	1:4		1:3		1:2		1:1		2:1		3:1		4:1		Mean			
		Len	Acc	Len	Acc	Len	Acc	Len	Acc	Len	Acc	Len	Acc	Len	Acc	Len	Acc		
Non-thinking	T-NT	271.3	0.737	245.9	0.692	254.8	0.700	246.8	0.640	316.6	0.695	295.1	0.667	328.8	0.676	279.9	0.687		
	NT-T	245.1	0.664	260.1	0.716	278.2	0.684	270.3	0.692	269.1	0.696	343.6	0.697	399.8	0.693	295.2	0.692		
	Mix	268.3	0.703	254.3	0.706	255.1	0.676	272.1	0.717	273.7	0.698	326.7	0.677	360.5	0.687	287.2	0.695		
	Mean	261.6	0.701	253.4	0.705	262.7	0.687	263.1	0.683	286.5	0.696	321.8	0.680	363.0	0.685	-	-		
Thinking	T-NT	5455.0	0.188	5632.5	0.198	5065.0	0.174	5519.8	0.186	6242.4	0.220	6858.5	0.224	7027.6	0.221	5971.5	0.202		
	NT-T	5731.7	0.200	5779.5	0.192	5938.7	0.204	5606.7	0.196	6225.7	0.212	6682.2	0.224	7080.8	0.227	6149.3	0.208		
	Mix	5877.1	0.208	5333.3	0.204	5438.8	0.190	5911.4	0.193	6211.7	0.215	7000.4	0.218	6888.5	0.225	6094.5	0.208		
	Mean	5687.9	0.199	5581.8	0.198	5480.8	0.189	5679.3	0.192	6226.6	0.216	6847.0	0.222	6999.0	0.224	-	-		

RQ2: Effect of Training Schedule

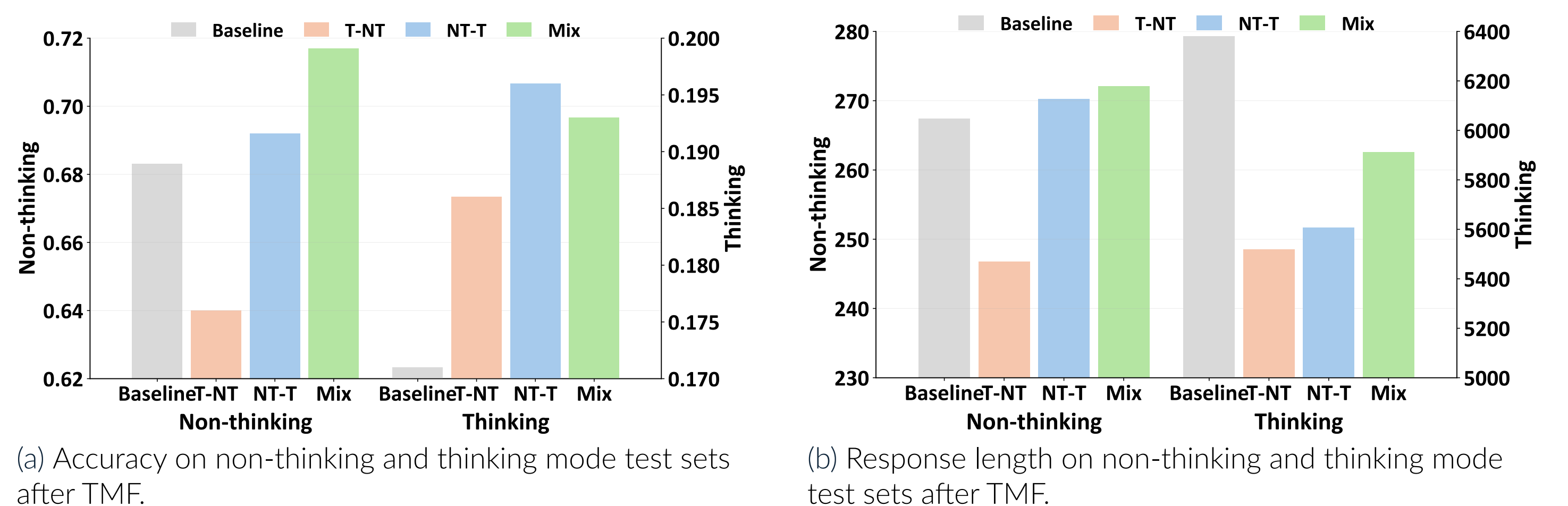


Figure. The effect of different training schedules under a fixed thinking-to-non-thinking data ratio of 1:1.

RQ3: Effect of Data Ratio

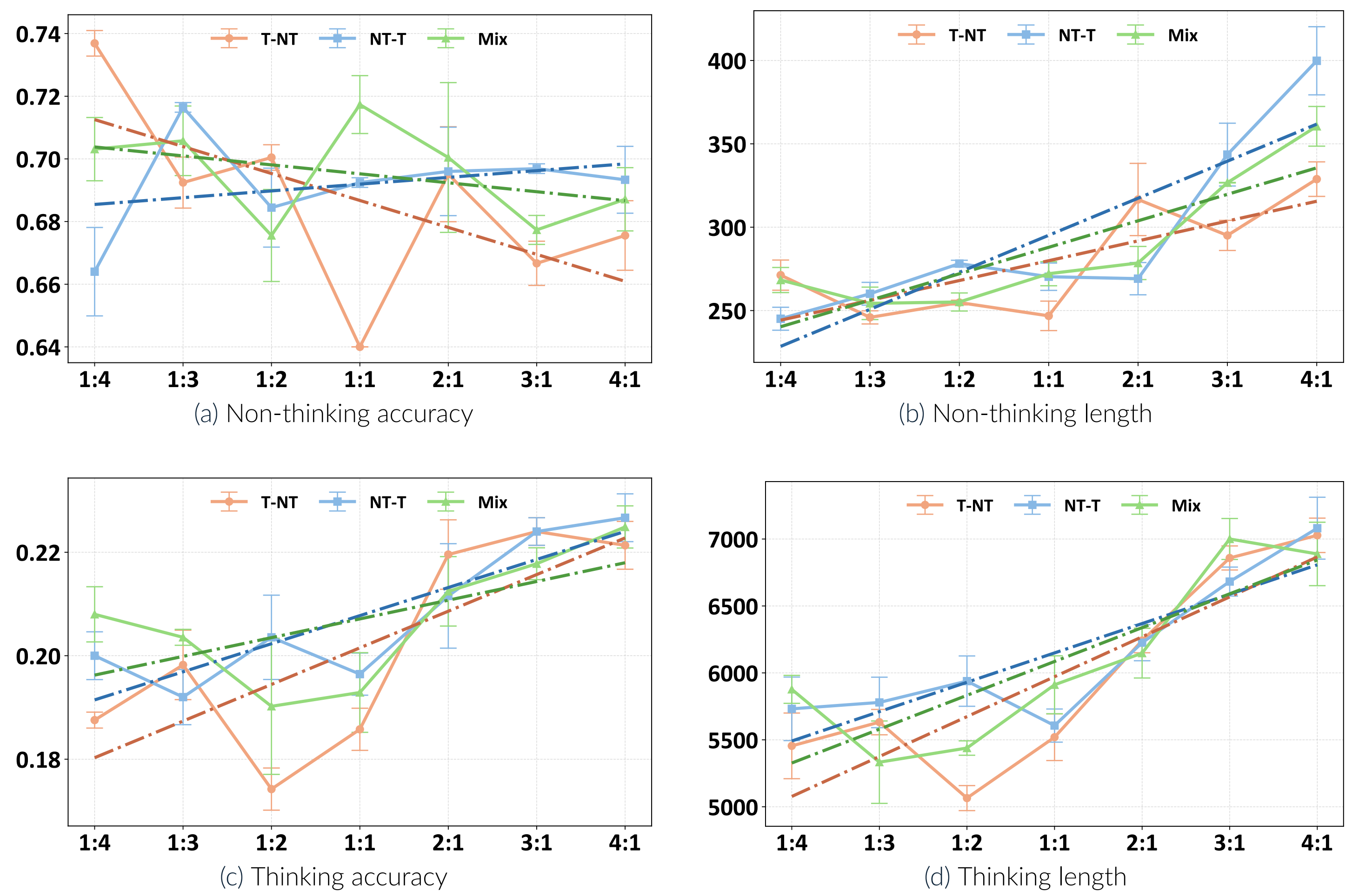


Figure. Performance comparison of accuracy and reasoning length under different data ratios and training schedules. Higher non-thinking ratios improve non-thinking accuracy but degrade thinking accuracy; under the Mix schedule, non-thinking accuracy increases with the data ratio, while thinking accuracy consistently increases across all schedules.

RQ4: Effect of Data Ratio

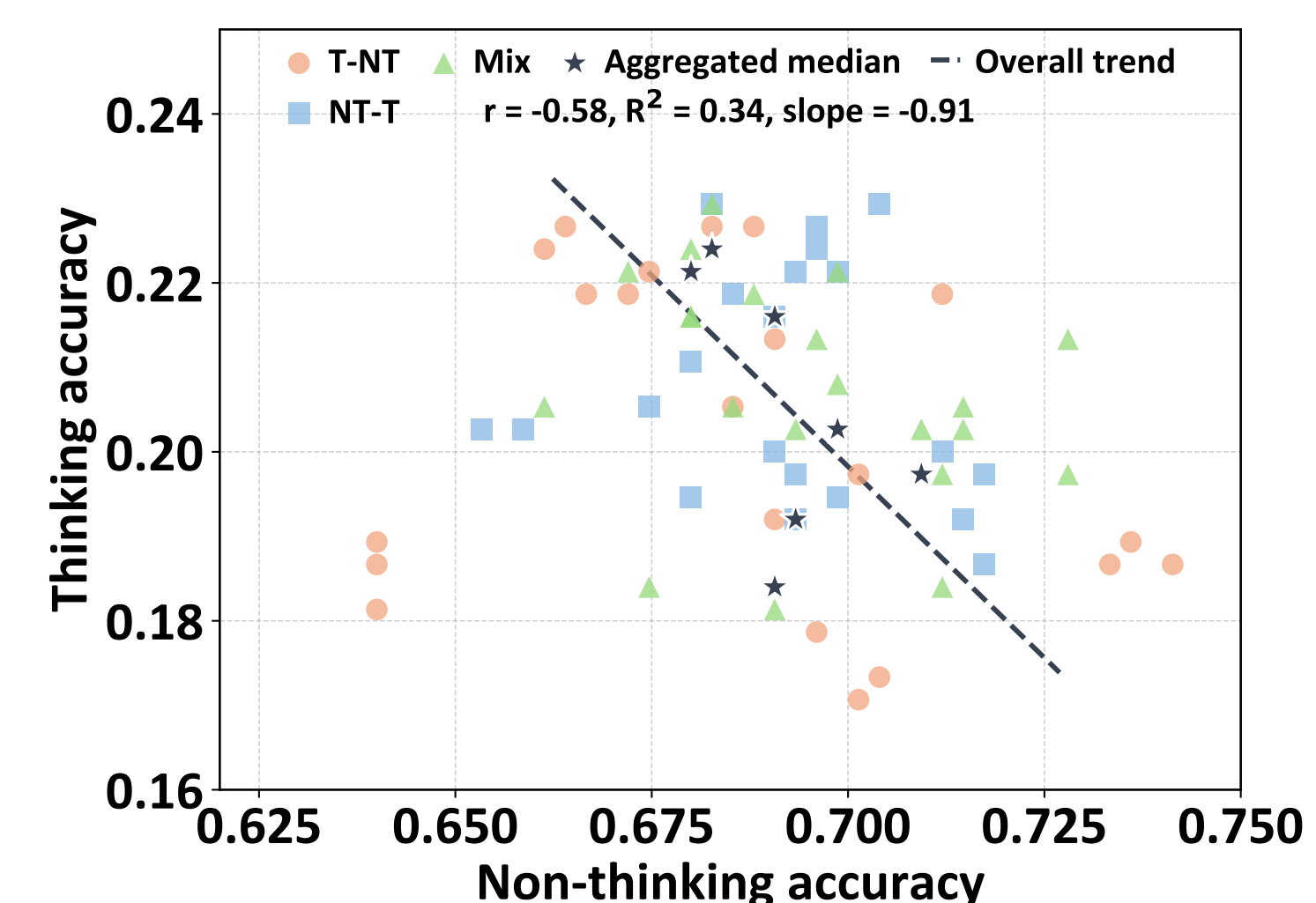


Figure. Scatter plot of different training schedules and data ratios. The dashed line indicates the fitted trend line of aggregation of the training schedules, showing a clear negative correlation ($r = -0.58$, $R^2 = 0.34$, slope = -0.91).

Take-home Messages

Interaction between two Mode: joint training shows asymmetric interference.

Data ratio controls the trade-off: more non-thinking data favors concise answers but harms long-form mathematical reasoning.

Schedule and ratio should be co-designed; interleaved Mix training provides the most robust overall performance.